ELEC3544 · Fall 2026

Data Science with Foundation Models

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The University of Hong Kong



Founding Members of ELEC3544!

Instructor

Invented and polished this course from 3 years ago!



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TAs and Students

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And ALL TAs and students from previous years!!!

> Thanks for starting and contributing to such a unique and visionary course!

Quick survey – Are you using AI agents?



Anthropic Fable 5



OpenAI GPT-5.6 Sol



Kimi K3



GLM 5.2



Google Gemini 3.7



Grok 4.5



Qwen 3.8 Max



DeepSeek V4 Pro



Muse Spark 1.2



ByteDance Doubao

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Why This Course, Why Now?

1. **A Convergence Moment**
   Powerful AI tools are now accessible to everyone. The barrier to entry has never been lower.

2. **First Generation of AI-Native Data Scientists**
   You are the first a few cohorts learning data science with AI tools from day one — not as an afterthought.

3. **The Rules Are Being Rewritten**
   Traditional workflows are transforming. Those who adapt thoughtfully will thrive.

> This course prepares you to think critically about **when** and **how** to leverage AI tools effectively.

The Evidence Is Clear: AI Is Transforming Data Science

TOP SKILLS IN JOB POSTINGS (2025)

Python	85%
Machine Learning	77%
SQL	59%
NLP Skills	5% → 19%

NLP demand nearly quadrupled due to LLM proliferation

NEW ROLES EMERGING

LLM Engineer
$123K - $224K

AI Engineer
Building apps on foundation models

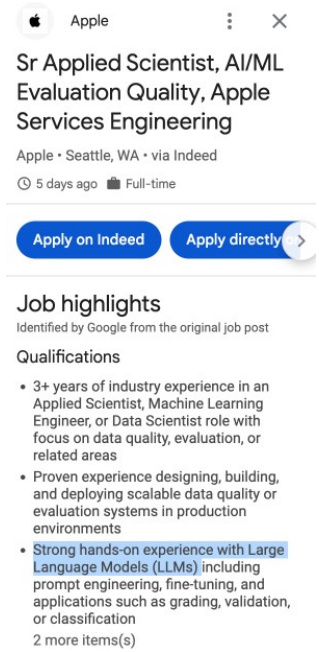
MLOps Engineer
Model deployment lifecycle

AI Product Manager
Bridging tech and business

MARKET REALITY

$152K
Entry-level salary (up $40K from 2024)

But experience requirements increasing. Employers want **fewer, more capable** practitioners.



The message: Data science is evolving rapidly. Those who combine strong fundamentals with AI fluency will thrive.